

We claim that

$$(1 + \Delta_{Dunkl})^{-\frac{1}{2}} M_\phi - (1 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}})^{-\frac{1}{2}} M_\phi \in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx))$$

if $\phi \in C_c^\infty(\mathbb{R}^N)$ is supported away from boundaries of the chambers.

We have

$$\begin{aligned} LHS &= \int_0^\infty \left(\frac{1}{1 + \lambda^2 + \Delta_{Dunkl}} - \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} \right) M_\phi d\lambda = \\ &= \pm \int_0^\infty \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} \left(M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}} - \Delta_{Dunkl} \right) \frac{1}{1 + \lambda^2 + \Delta_{Dunkl}} M_\phi d\lambda = \\ &= \pm \int_0^\infty \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} \left(M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}} - \Delta_{Dunkl} \right) M_\phi \frac{1}{1 + \lambda^2 + \Delta_{Dunkl}} d\lambda \pm \\ &\pm \int_0^\infty \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} \left(M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}} - \Delta_{Dunkl} \right) \frac{1}{1 + \lambda^2 + \Delta_{Dunkl}} [\Delta_{Dunkl}, M_\phi] \frac{1}{1 + \lambda^2 + \Delta_{Dunkl}} d\lambda = \\ &= \pm \int_0^\infty \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} \left(M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}} - \Delta_{Dunkl} \right) M_\phi \frac{1}{1 + \lambda^2 + \Delta_{Dunkl}} d\lambda \pm \\ &\quad \pm \int_0^\infty \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} [\Delta_{Dunkl}, M_\phi] \frac{1}{1 + \lambda^2 + \Delta_{Dunkl}} d\lambda \pm \\ &\quad \pm \int_0^\infty \frac{1}{1 + \lambda^2 + \Delta_{Dunkl}} [\Delta_{Dunkl}, M_\phi] \frac{1}{1 + \lambda^2 + \Delta_{Dunkl}} d\lambda. \end{aligned}$$

The operator

$$\left(M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}} - \Delta_{Dunkl} \right) M_\phi$$

is of the shape

$$P + \sum_{v \in R} U_{\sigma_v} P_k,$$

where P and P_k are differential operators of degree 1 with smooth compactly (and away from the boundaries) supported coefficients. So is the operators

$$[\Delta_{Dunkl}, M_\phi].$$

It now remains to show that

$$\begin{aligned} \int_0^\infty \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} P \frac{1}{1 + \lambda^2 + \Delta_{Dunkl}} d\lambda &\in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)), \\ \int_0^\infty \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} U_g P \frac{1}{1 + \lambda^2 + \Delta_{Dunkl}} d\lambda &\in (\mathcal{L}_{N,\infty})_0(L_2(\mathbb{R}^N, w(x)dx)), \end{aligned}$$

for every differential operator P of degree 1 with smooth compactly (and away from the boundaries) supported coefficients and for every $g \in G$.

Since M_w and Δ commute with U_g , it follows that it is sufficient to prove only the first assertion. Fix $\psi \in C_c^\infty(\mathbb{R}^N)$ such that $P = PM_\psi$. We write

$$\begin{aligned} &\frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} P \frac{1}{1 + \lambda^2 + \Delta_{Dunkl}} = \\ &= \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} P \cdot M_\psi \frac{1}{1 + \lambda^2 + \Delta_{Dunkl}}. \end{aligned}$$

Both factors are in $\mathcal{L}_{N,\infty}$ (this requires SOME SORT of Cwikel estimates). Hence, the product is in $\mathcal{L}_{\frac{9N}{10},\infty}$. We now have

$$\begin{aligned} & \left\| \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} P \frac{1}{1 + \lambda^2 + \Delta_{Dunkl}} \right\|_{\frac{9N}{10},\infty} \leq \\ & \leq \left\| \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} P \right\|_{\frac{9N}{5},\infty} \times \\ & \quad \times \left\| M_\psi \frac{1}{1 + \lambda^2 + \Delta_{Dunkl}} \right\|_{\frac{9N}{5},\infty}. \end{aligned}$$

Obviously,

$$\begin{aligned} & \left\| \frac{1}{1 + \lambda^2 + M_{w^{-\frac{1}{2}}} \Delta M_{w^{\frac{1}{2}}}} P \right\|_{\frac{9N}{5},\infty} = \left\| \frac{1}{1 + \lambda^2 + \Delta} M_{w^{\frac{1}{2}}} P M_{w^{-\frac{1}{2}}} \right\|_{\mathcal{L}_{\frac{9N}{5},\infty}(\mathbb{R}^N)} \leq \\ & \leq c_P \|(1 + \lambda^2 + \Delta)^{-\frac{1}{2}} M_\theta\|_{\mathcal{L}_{\frac{9N}{5},\infty}(\mathbb{R}^N)} = \\ & = c_P \|(1 + \lambda^2 + \Delta)^{-\frac{2}{9}} \cdot (1 + \lambda^2 + \Delta)^{-\frac{5}{18}} M_\theta\|_{\mathcal{L}_{\frac{9N}{5},\infty}(\mathbb{R}^N)} \leq \\ & = c_P (1 + \lambda^2)^{-\frac{2}{9}} \|(1 + \Delta)^{-\frac{5}{18}} M_\theta\|_{\mathcal{L}_{\frac{9N}{5},\infty}(\mathbb{R}^N)}. \end{aligned}$$

Similarly,

$$\begin{aligned} & \left\| M_\psi \frac{1}{1 + \lambda^2 + \Delta_{Dunkl}} \right\|_{\frac{9N}{5},\infty} = \\ & = \|M_\psi (1 + \lambda^2 + \Delta_{Dunkl})^{-\frac{5}{18}} \cdot (1 + \lambda^2 + \Delta_{Dunkl})^{-\frac{13}{18}}\|_{\frac{9N}{5},\infty} \leq \\ & \leq (1 + \lambda^2)^{-\frac{13}{18}} \|M_\psi (1 + \Delta_{Dunkl})^{-\frac{5}{18}}\|_{\frac{9N}{5},\infty}. \end{aligned}$$

So, the integrand is controlled by

$$(1 + \lambda^2)^{-\frac{17}{18}}.$$